

Unit 6: Assemblies, Mechanisms, and Aerospace Systems

Quiz 6A: Assemblies, Components, Joints, and Degrees of Freedom

Name: _____ Period: _____

Date: _____ Score: _____

Instructions

Answer each question independently. Choose the best answer for multiple choice, complete the matching section, and respond to short-response prompts in complete sentences.

Score:

/ 25

Part A: Multiple Choice

1. An assembly is...
A. Multiple parts working together
B. A single note
C. Only a sketch
D. Only a quiz
2. A component is...
A. A distinct part within a design or assembly
B. A random color
C. A page number
D. A type of cleanup
3. A joint defines...
A. How parts connect or move relative to each other
B. Only the font
C. The final score
D. Where the logo goes
4. Degrees of freedom describe...
A. The ways a part can move
B. How many students are absent
C. The number of sketches
D. Only material thickness
5. A rigid joint usually...
A. Locks parts together with no relative motion
B. Makes parts spin freely
C. Deletes parts
D. Measures mass
6. A revolute joint allows...
A. Rotation around an axis
B. Only sliding
C. Only scaling
D. No motion ever
7. Why plan assemblies before modeling?
A. To understand part relationships and motion
B. To avoid saving
C. To skip dimensions
D. To make fewer notes
8. A poorly defined joint can cause...

- A. Incorrect motion or assembly errors
- C. No consequences

- B. Better documentation automatically
- D. Perfect function

Part B: Vocabulary / Matching

Write the letter of the best matching term in the blank.

- | | |
|--------------------------------|---|
| _____ Assembly | A. A group of parts that fit or work together. |
| _____ Component | B. A distinct part within an assembly. |
| _____ Joint | C. A connection defining how parts relate or move. |
| _____ Degree of freedom | D. A possible direction or type of motion. |
| _____ Rigid joint | E. A joint that prevents relative motion between parts. |

Part C: Short Response

1. Explain why an assembly needs more planning than a single part.

2. Describe the difference between a rigid joint and a revolute joint.

Unit 6: Assemblies, Mechanisms, and Aerospace Systems

Quiz 6B: Mechanisms, Motion, Control Surfaces, Landing Gear, and Payload Doors

Name: _____ Period: _____

Date: _____ Score: _____

Instructions

Answer each question independently. Choose the best answer for multiple choice, complete the matching section, and respond to short-response prompts in complete sentences.

Score:

/ 25

Part A: Multiple Choice

- A mechanism is...
 - A. A system of parts that creates or controls motion
 - B. Only a material list
 - C. Only a drawing title
 - D. A static worksheet
- A control surface on an aircraft helps...
 - A. Change or control motion through air
 - B. Store pencils
 - C. Grade presentations
 - D. Make sketches darker
- Landing gear must often balance...
 - A. Strength, weight, motion, and reliability
 - B. Only color
 - C. Only page number
 - D. Only class time
- A payload door mechanism should...
 - A. Open/close reliably while protecting the payload
 - B. Use random motion
 - C. Ignore constraints
 - D. Avoid documentation
- Which joint may be useful for a hinge?
 - A. Revolute
 - B. Fixed text
 - C. Material
 - D. Scale only
- Why test mechanism motion?
 - A. Parts may interfere, bind, or move incorrectly
 - B. Testing is optional
 - C. Motion is always perfect in sketches
 - D. It replaces CAD
- Interference means...
 - A. Parts occupy the same space or collide when assembled
 - B. Parts are labeled clearly
 - C. A successful presentation
 - D. A material property only

8. Which is a good mechanism design question?
- A. What motion is required and what limits that motion? B. How can we avoid all sketches?
- C. What color is the rubric? D. Can we ignore the payload?

Part B: Vocabulary / Matching

Write the letter of the best matching term in the blank.

- | | |
|------------------------------|--|
| _____ Mechanism | A. A system of parts that creates or controls motion. |
| _____ Control surface | B. A movable aircraft surface used to control flight behavior. |
| _____ Hinge | C. A joint that allows rotation. |
| _____ Interference | D. A collision or overlap between parts in an assembly. |
| _____ Range of motion | E. The amount or limits of movement a part can make. |

Part C: Short Response

1. Describe one aerospace mechanism and the motion it must create.

2. Explain why interference checks matter in a moving assembly.

Unit 6: Assemblies, Mechanisms, and Aerospace Systems

Quiz 6C: Exploded Views, Assembly Instructions, BOMs, and System Documentation

Name: _____ Period: _____

Date: _____ Score: _____

Instructions

Answer each question independently. Choose the best answer for multiple choice, complete the matching section, and respond to short-response prompts in complete sentences.

Score:

/ 25

Part A: Multiple Choice

- An exploded view shows...
 - How parts fit together by spacing them apart visually
 - Only one hidden line
 - Only final cost
 - A broken design only
- Assembly instructions should be...
 - Clear, ordered, and specific
 - Random
 - Only verbal
 - Unlabeled
- A system-level BOM should include...
 - All parts, quantities, materials, and notes
 - Only the largest part
 - Only team names
 - Only sketches
- Why document assembly steps?
 - So another person can build or inspect the system
 - To make the packet longer
 - To hide mistakes
 - To avoid revisions
- Which is a useful assembly instruction?
 - Insert the 1/8 in axle through the hinge holes and secure with two collars
 - Put it together
 - Do the thing
 - Make it work
- A subassembly is...
 - A smaller assembly within a larger system
 - A missing part
 - A title page
 - A sketch error
- Why include orientation notes?
 - Parts may fit only in a specific direction
 - Orientation never matters
 - It only changes color
 - It replaces dimensions

8. Which document helps communicate how motion should work?
- A. Assembly documentation with joint/motion notes B. A blank page
- C. Only a team photo D. A random title

Part B: Vocabulary / Matching

Write the letter of the best matching term in the blank.

- | | |
|-----------------------------------|--|
| _____ Exploded view | A. A drawing showing parts separated to communicate how they fit together. |
| _____ Assembly instruction | B. A written step explaining how to build or connect parts. |
| _____ Subassembly | C. A smaller group of assembled parts within a larger assembly. |
| _____ Orientation note | D. A note explaining direction or placement of a part. |
| _____ System documentation | E. Documents that explain parts, assembly, function, and motion. |

Part C: Short Response

1. Explain why exploded views are helpful for assembly documentation.

2. Write one clear assembly instruction for a hinged payload door.
